

A Memoir on the Transformation of Elliptic Functions (continued from p. 142): Articles 34–52

Arthur Cayley
Collected Mathematical Papers, Vol. IX, Paper 578
pages 143–155

The present file covers PDF pages 163–175 of the scanned volume, corresponding to book pages 143–155. It is a sub-chunk of the requested range 163–187 (PDF pages 176–187 deferred owing to the dense filter-block analysis of §§52–71, which warrants a separate file).

[Book p. 143]

34. Next $n = 5$. Here, starting from $\frac{1}{M} = \frac{v - u^3}{v(1 - uv^3)}$, the changes give

$$\frac{1}{M} = \frac{v - u^3}{v(1 - uv^3)}, \quad 5M = \frac{u + v^3}{v(1 + u^3v)}, \quad \frac{v^3}{u^3M} = 1 + \frac{2v}{u^3}, \quad \frac{u^3}{v^3} 5M = 1 - \frac{2u}{v^3};$$

viz. the third and the fourth forms agree with the first and the second forms respectively; that is, there are only two independent forms, and the elimination of M from these gives

$$5uv(1 - uv^3)(1 + u^3v) - (v - u^3)(u + v^3) = 0,$$

which is a form of the modular equation.

35. In the case $n = 7$, starting from $\frac{1}{M} = \frac{-7u(1 - uv)(1 - uv + u^2v^2)}{u - v^3}$ (as to this see *post*, No. 43), the forms are

$$\frac{1}{M} = \frac{-7u(1 - uv)(1 - uv + u^2v^2)}{u - v^3} \quad \dots (1),$$

$$-7M = \frac{-7v(1 - uv)(1 - uv + u^2v^2)}{v - u^3} \quad \dots (2),$$

$$\frac{v^3}{u^3M} = \frac{-7v^3(1 - uv)(1 - uv + u^2v^2)}{u^3(v - u^3)} \quad \dots (3),$$

$$-\frac{u^3}{v^3} 7M = \frac{-7u^3(1 - uv)(1 - uv + u^2v^2)}{v^3(v - u^3)} \quad \dots (4);$$

so that here again the third and the fourth forms are identical with the second and the third forms respectively; there are thus only two forms, and the elimination of M gives

$$(u - v^3)(v - u^3) - 4uv(1 - uv)(1 - uv + u^2v^2)^2 = 0,$$

which is a form of the modular equation.

[Book p. 144]

36. If in the foregoing equation

$$F'v \cdot \frac{1}{M} = R(u, v),$$

we make the change $u, v, \frac{1}{M}$ into $v, \pm u, \pm nM$, it becomes

$$\pm F'u \cdot nM = R(v, \pm u);$$

combining these equations, we have

$$\pm nM^2 \cdot \frac{F'u}{F'v} = \frac{R(v, \pm u)}{R(u, v)};$$

or, substituting herein the foregoing value

$$M^2 = -\frac{1}{n} \frac{(1-v^3)vF'v}{(1-u^3)uF'u},$$

this becomes

$$\mp \frac{v(1-v^3)}{u(1-u^3)} = \frac{R(v, \pm u)}{R(u, v)} \quad \begin{array}{l} + \text{ for } n \equiv 3 \text{ or } 5 \pmod{8}, \\ - \text{ for } n \equiv 1 \text{ or } 7 \pmod{8}, \end{array}$$

which must agree with the modular equation. Thus in the last-mentioned case $n = 3$, we have

$$\frac{1}{3} F'v \cdot \frac{1}{M} = 3(v^2u^2 + 2u^3v + 1)u,$$

or, say

$$R(u, v) = (v^2u^2 + 2u^3v + 1)u,$$

and therefore

$$R(v, -u) = (v^2u^2 - 2uv^3 + 1)v;$$

the equation is

$$+ \frac{v(1-v^3)}{u(1-u^3)} = \frac{(v^2u^2 - 2uv^3 + 1)v}{(v^2u^2 + 2u^3v + 1)u},$$

which is right; because Jacobi, p. 82, [*Ges. Werke*, t. I, p. 137], for the modular equation, gives

$$1 - u^3 = (1 - u^3v^3)(v^2u^2 + 2u^3v + 1), \quad 1 - v^3 = (1 - u^3v^3)(v^2u^2 - 2uv^3 + 1).$$

Observe that the general equation

$$\mp \frac{v(1-v^3)}{u(1-u^3)} = \frac{R(v, \pm u)}{R(u, v)}$$

no longer contains the functions $F'v$, $F'u$, which enter into Jacobi's expression of M^2 .

Theorem in connexion with the multiplication of elliptic functions. Art. Nos. 37 to 40.

37. The theory of multiplication gives an important theorem in regard to transformation. Starting with the white transformation

$$\frac{1-y}{1+y} = \frac{1-x}{1+x} \frac{(\alpha' - \beta x + \gamma x^2 - \dots)^2}{(\alpha + \beta x + \gamma x^2 + \dots)^2}, \quad = \frac{1-x}{1+x} \left(\frac{P - Qx}{P + Qx} \right)^2,$$

we may form a like transformation,

$$\frac{1-z}{1+z} = \frac{1-y}{1+y} \frac{(\alpha' - \beta'y + \gamma'y^2 - \dots)^2}{(\alpha + \beta'y + \gamma'y^2 + \dots)^2}, \quad = \frac{1-y}{1+y} \left(\frac{P' - Q'y}{P' + Q'y} \right)^2,$$

such that the combination of the two gives a multiplication, viz. for the relation between y , z , deriving w from z as v from u , we have $w = u$; and instead of M we have M' , $= \pm \frac{1}{nM}$; that is, we have

$$\frac{dx}{\sqrt{1-x^2 \cdot 1-u^2x^2}} = \frac{Mdy}{\sqrt{1-y^2 \cdot 1-v^2y^2}},$$

$$\frac{dy}{\sqrt{1-y^2 \cdot 1-v^2y^2}} = \frac{M'dz}{\sqrt{1-z^2 \cdot 1-w^2z^2}},$$

and thence

$$\frac{dx}{\sqrt{1-x^2 \cdot 1-u^2x^2}} = \frac{\pm \frac{1}{n} dz}{\sqrt{1-z^2 \cdot 1-u^2z^2}};$$

or, writing $x = \sin \theta$, we have $z = \pm \sin n\theta$; $\pm$ is here $(-)^{\frac{n-1}{2}}$, viz. it is $-$ for $n \equiv 3$ or $7 \pmod{8}$, and $+$ for $n \equiv 1$ or $5 \pmod{8}$.

Now in part effecting the substitution, we have

$$\frac{1-z}{1+z} = \frac{1-x}{1+x} \left(\frac{P-Qx}{P+Qx} \right)^2 \left(\frac{P'-Q'y}{P'+Q'y} \right)^2,$$

where y denotes its value in terms of x .

And from the theory of elliptic functions, replacing $\sin n\theta$, $\sin \theta$ by their values $\pm z$, x , we have an equation

$$\frac{1-z}{1+z} = \frac{1-x}{1+x} \frac{(A-Bx+Cx^2-\dots)^2}{(A+Bx+Cx^2+\dots)^2},$$

where $A-Bx+Cx^2-\dots$, $A+Bx+Cx^2+\dots$ are given functions each of the order $\frac{1}{2}(n^2-1)$; viz. the coefficients are given functions of k , or, what is the same thing, of u . Comparing the two results, we see that in the sthic transformation the sought-for function, $\alpha + \beta x + \gamma x^2 + \dots$ of the order $\frac{1}{2}(n-1)$, is a factor of a given function $A+Bx+Cx^2+\dots$ of the order $\frac{1}{2}(n^2-1)$.

38. Considering the modular equation as known, then by what precedes we have

$$\alpha + \beta x + \gamma x^2 + \dots = \alpha \left\{ 1 + \frac{\beta}{\alpha} x + \dots + \frac{u^m}{v} x^{\frac{1}{2}(n-1)} \right\};$$

that is, the given function $A+Bx+Cx^2+\dots$ has a factor $1 + \frac{\beta}{\alpha} x + \dots + \frac{u^m}{v} x^{\frac{1}{2}(n-1)}$, of which one (the last) coefficient $\frac{u^m}{v}$ is known, and we are hence able theoretically to

[Book p. 145]

determine all the other coefficients rationally in terms of u , v ; that is, the modular equation being known, we can theoretically complete the solution of the transformation problem. I do not, however, see the way to obtaining a convenient solution in this manner.

39. The formula in question for $n = 3$ is

$$\frac{1 + \sin 3\theta}{1 - \sin 3\theta} = \frac{1 - \sin \theta}{1 + \sin \theta} \frac{(1 + 2 \sin \theta - 2k^2 \sin^3 \theta - k^2 \sin^4 \theta)^2}{(1 - 2 \sin \theta + 2k^2 \sin^3 \theta - k^2 \sin^4 \theta)^2},$$

which, putting therein $x = \sin \theta$, $z = -\sin 3\theta$, and replacing k by u^3 , may be written

$$1 + z(+) = (1+x)(1-2x+2u^3x^3-u^4x^4)^2(+),$$

where the signs $(+)$ indicate denominators which are obtained from the numerators by changing the signs of x , z respectively.

The theorem in regard to $n = 3$ thus is, $1 + \frac{u^3}{v} x$ is a factor of $1 - 2x + 2u^3x^3 - u^4x^4$; viz. writing in the last-mentioned function $x = -\frac{v}{u^3}$, we ought to have

$$0 = 1 + 2 \frac{v}{u^3} - 2 \frac{v^3}{u^6} - \frac{v^4}{u^4},$$

that is,

$$u^4 + 2uv - 2u^3v^3 - v^4 = 0,$$

which is in fact the modular equation.

40. And so for $n = 5$, if $x = \sin \theta$, $z = \sin 5\theta$; and for $n = 7$, if $x = \sin \theta$, $z = -\sin 7\theta$; the formulas are:—

$$\begin{array}{c|c}
\begin{array}{l}
n = 5, \\
1 + z = (1 + x) \left[\begin{array}{l} 1 \\ (+) + 2x \\ - 4x^2 \\ - 10x^3 \\ + 5x^4 \\ + 4v^4(3 + 2u^3)x^5 \\ + 4v^4(1 - u^3)x^6 \\ - 4v^4(2 + 5u^3)x^7 \\ - 3u^4x^7 \\ - 4u^3(2 + 5u^3)x^9 \\ + 10u^{10}x^{10} \\ + 4u^{14}(3 + 2u^3)x^{11} \\ + 5u^4x^{12} \\ - 2u^{14}x^{13} \\ - u^{20}x^{14} \end{array} \right]
\end{array}
&
\begin{array}{l}
n = 7, \\
1 + z = (1 + x) \left[\begin{array}{l} 1 \\ (+) - 4x \\ + (2 + 7x^2)x^2 \\ - 14u^3x^4 \\ + 8u^3(16 + 25u^4 + 4u^6)x^5 \\ - 8u^3(16 + 30u^4 + 4u^6)x^6 \\ - u^4(16 + 30u^4 + 14u^6)x^7 \\ + 4u^3(8 + 51u^3 + 15u^6)x^8 \\ + 2u^4(4 + u^3)x^9 \\ - u^4(144 + 303u^3 + 16u^6)x^{10} \\ - 9u^4(4 + 25u^3 + 10u^6)x^{11} \\ - 8u^3(4 + 57u^3 + 46u^6)x^{12} \\ + 56u^{10}(2 + u^3)x^{13} \\ - 4u^{20}x^{14} \\ - 4u^{20}x^{15} \\ + u^{20}x^{16} \\ (+) \end{array} \right]
\end{array}
\end{array}$$

[Book p. 147]

Term in [] has factor

$$1 + \frac{\beta}{\alpha}x + \frac{u^2}{v}x^2;$$

$u = 1$, term in [] is $(1 + x)^4(1 - x)^2$.

Term in [] has factor

$$1 + \frac{\beta}{\alpha}x + \frac{\gamma}{\alpha}x^2 + \frac{u^3}{v}x^3;$$

$u = 1$, term in [] is $(1 + x)^6(1 - x)^{10}$.

The transformations $n = 3, 5, 7, 11$. Art. Nos. 41 to 51.

41. The cubic transformation, $n = 3$.

I reproduce the results already obtained; since there are only two coefficients α, β , these are also the last but one and last coefficients ρ, σ . Hence, from the values of $\alpha, \beta, \rho, \sigma$, we have

$$\alpha = 1, \quad 2\alpha = \frac{v^2}{u} \left(\frac{1}{M} - \frac{u^3}{v^3} \right),$$

$$2\beta = \frac{1}{M} - 1, \quad \beta = \frac{u^3}{v};$$

the two values of $\frac{1}{M}$ are thus $\frac{1}{M} = 2 \frac{u^3}{v} + \frac{v^3}{u^3} = 1 + \frac{2u^3}{v}$, giving the modular equation

$$v^4 + 2v^3u^3 - 2uv - u^4 = 0,$$

and we then have

$$\frac{1 - y}{1 + y} = \frac{1 - x}{1 + x} \left(\frac{v - u^3x^2}{v + u^3x^2} \right)^2.$$

[Book p. 148]

42. The quintic transformation, $n = 5$.

Here there are the three coefficients α, β, γ , or β, γ are the last but one and last coefficients ρ, σ ; we have

$$\alpha = 1, \quad 2\beta = v^2u \left(\frac{1}{M} - \frac{u^5}{v^5} \right),$$

$$2\beta = \frac{1}{M} - 1, \quad \gamma = \frac{u^2}{v}.$$

Comparing the two values of β , we have

$$\frac{1}{M} = \frac{v - u^5}{v(1 - v^4u)}, \quad \text{and then} \quad \alpha = 1, \quad 2\beta = \frac{u(v^2 - u^4)}{v(1 - v^4u)}, \quad \gamma = \frac{u^2}{v},$$

so that only the modular equation remains to be determined.

The unused equation is

$$2\alpha\gamma + 2\alpha\beta + \beta^2 = \frac{v^2}{u^2} (2\alpha\gamma + 2\beta\gamma + \beta^2),$$

which, putting therein $\alpha = 1$, may be written

$$2uv^2(u^2 - v^2) = 2\beta(\gamma v^2 - u^2);$$

attending to the value of β , this divides by $v^2 - u^2$; in fact the equation may be written

$$2v + \beta^2 = -\frac{u}{v(v^2 - u^4)}(\gamma v^2 - u^2);$$

and then completing the substitution, and integralizing, this becomes

$$[8m^2(1 - v^4u)^2 + (v^2 - u^4)^2] = 4uv(u^4 + v^2)(1 - u^4v)(1 - uv^4),$$

viz. this is

$$4(1 - v^4u)uv[2u^4(1 - v^4u) - (u^4 + v^2)(1 - uv^4)] + (v^2 - u^4)^2 = 0;$$

and the term in [] being $-(v^2 - u^4)(1 + uv^3)$, the whole again divides by $v^2 - u^4$, and the equation thus becomes

$$(v^2 + u^4)(v^2 - u^4) - 4uv(1 - v^4u)(1 + uv^3) = 0,$$

which is the modular equation.

43. The septic transformation, $n = 7$.

I do not propose to complete the solution directly from the fundamental equations for $\alpha, \beta, \gamma, \delta$, but resort to the known modular equation, and to an expression of M which I obtain by means thereof.

The modular equation is

$$(1 - u^7)(1 - v^7) - (1 - uv)^8 = 0,$$

[Book p. 149]

which may also be written

$$(v - u^7)(u - v^7) + 7uv(1 - uv)(1 - uv + u^2v^2)^2 = 0,$$

as can be at once verified; but it also follows from Cauchy's identity

$$(x + y)^7 - x^7 - y^7 = 7xy(x + y)(x^2 + xy + y^2)^2.$$

We then have

$$\begin{aligned} M &= -\frac{1}{n} \frac{(1 - v^4)vF'v}{(1 - u^4)uF'u} = -\frac{v^4}{1 - u^4}(1 - uv)^2, \\ &= -\frac{u^4}{1 - u^4}u(1 - v^4)^2. \end{aligned}$$

Moreover

$$\begin{aligned} uF'u &= -u^4(1 - v^4) + uv(1 - uv)^2 = -\frac{u^4}{1 - u^4}(1 - uv)^2 + uv(1 - uv)^2 \\ &= \frac{(1 - uv)^2}{1 - u^4}u(v - u^7); \end{aligned}$$

and similarly

$$F'v = \frac{(1 - uv)^2}{1 - v^4}v(u - v^7);$$

whence

$$\frac{1}{M} = \frac{-7u(v - u^7)}{u - v^7}.$$

Writing this under the form

$$\frac{1}{M^4} = \frac{-7uv(v - u^7)(u - v^7)}{(u - v^7)^2} = \frac{49u^2(1 - uv)^4(1 - uv + u^2v^2)^4}{(u - v^7)^2},$$

I find, as will appear, that the root must be taken with the sign $-$, and that we thus have

$$\frac{1}{M} = \frac{7u(1 - uv)(1 - uv + u^2v^2)}{u - v^7}, \quad \text{whence also } \frac{M}{1} = \frac{u(1 - uv)(1 - uv + u^2v^2)}{u - v^7}.$$

44. Recurring now to the fundamental equations for the septic transformation, the coefficients are $\alpha, \beta, \gamma, \delta$, and we have

$$\alpha = 1, \quad 2\gamma = u^3v^2\left(\frac{1}{M} - \frac{u^7}{v^7}\right),$$

$$2\beta = \frac{1}{M} - 1, \quad \delta = \frac{u^3}{v};$$

so that the coefficients are all given in terms of v, M . The unused equations are two equations, viz.

$$u^4(2\alpha\gamma + 2\alpha\beta + \beta^2) = v^4(\gamma^2 + 2\beta\gamma + 2\delta\gamma),$$

$$u^{-4}(\gamma^2 + 2\beta\gamma + 2\delta\gamma) = v^4(2\alpha\gamma + 2\beta\delta + 2\alpha\delta + \beta^2),$$

which, substituting therein for $\alpha, \beta, \gamma, \delta$ the foregoing values, give two equations; from these, eliminating M , we should obtain the modular equation, and then M in terms of u, v .

[Book p. 150]

Substituting in the first instance for α, δ their values, the equations are

$$u^4(2\beta + 2\beta\gamma + \beta^2) = v^4\left\{\gamma^2 + 2\frac{u^3}{v}(\beta + \gamma)\right\},$$

$$\gamma^2 + 2\beta\gamma + (2 + 2\beta)\frac{u^3}{v} = u^4v^2\left\{2\gamma + 2\beta\gamma + 2\frac{u^3}{v}\beta^2\right\}.$$

The first of these is

$$4(1 - uv)(2\beta + 2\gamma) + 4\beta - \frac{4u^3}{v}\gamma^2 = 0,$$

viz. this is

$$4(1 - uv)\left(\frac{1}{M} - 1 + \frac{u^3}{v^2}\left(\frac{1}{M} - \frac{u^7}{v^7}\right)\right) + \left(\frac{1}{M} - 1\right)^2 - \frac{4u^3}{v}\left(\frac{1}{M}\right)^2 = 0;$$

or, observing that in this equation the coefficient of $\frac{2}{M}$ is

$$(1 - u^4v^2)[2 - 2uv + 2uv^2 - \frac{u^3}{v}] - \frac{u^3}{v}\left(\frac{1}{M} - 1\right)^2,$$

$= (1 - u^4v^2)(1 - uv)^4 = (1 - uv)^4(1 + uv)$; the equation becomes

$$(1 - uv)^2\left[2 - 2uv + 2uv^2 - \frac{u^3}{v}\right]\left(1 + \frac{u^3}{v}\right) + (1 - uv)\left(1 + \frac{u^3}{v}\right) = 0.$$

45. This should be satisfied identically by the foregoing value of $\frac{1}{M}$; viz. it should be satisfied on writing therein

$$\frac{1}{M} = -\frac{7u}{v} \frac{v - u^7}{u - v^7}, \quad \frac{1}{M} = \frac{7u(1 - uv)(1 - uv + u^2v^2)}{u - v^7};$$

that is, we should have

$$-7\frac{u}{v}(v - u^7)(1 - uv^2) - 14uv(1 - uv)(1 + u^4v^4) + (u - v^7)\left\{1 - u^7 - 4(1 - uv)\left(1 + \frac{u^3}{v}\right)\right\} = 0,$$

where observe that the $-$ sign of the second term in the sign of the foregoing value of $\frac{1}{M}$; so that the identity being verified, it follows that the correct sign has been attributed to the value of $\frac{1}{M}$.

46. Multiplying by v , the equation is

$$-7(1-u^7)(1-v^7) - 14uv(1-uv)^2(1+u^4v^4) + [1-v^7][-8(1-uv) + 1-u^7] + 4(1-uv)(v-v^7)(u-v^7) = 0,$$

[Book p. 151]

viz. this is

$$-7(1-u^7)(1-v^7) + 7(1-uv)(1-v^7) - 14uv(1-uv)^2(1+u^4v^4) + (1-u^7)(1-v^7) - 8(1-uv)(1-v^7) + 4(1-uv)^2(u-v^7)(v-v^7) = 0.$$

In the second column the coefficient of $1-uv$ is $2-u^7-v^7$, viz. this is

$$= (1-u^7)(1-v^7) + 1-u^7v^7 = (1-u^7v^7) + (1-uv)^8.$$

Reducing also the other two columns by means of the modular equation, the equation thus becomes

$$-6(1-uv)^7 - (1-uv)^8 - 14uv(1-uv)^2(1+u^4v^4) + 8(1-uv)^9 - 28uv(1-uv)(1-uv+u^2v^2)^2 = 0.$$

This is in fact an identity; to show it, writing for convenience θ in place of uv , and observing that the terms

$$\begin{aligned} -(1-\theta)^8 + 8(1-\theta)^9 &= -(1-\theta)^8[8(1-\theta+\theta^2+\theta^3+\theta^4+\theta^5+\theta^6+\theta^7)] \\ &= (1-\theta)^9(7+6\theta+5\theta^2+4\theta^3+3\theta^4+2\theta^5+\theta^6), \end{aligned}$$

the whole equation divides by $(1-\theta)^7$; or throwing out this factor, it is

$$-6(1-\theta) - (1-\theta)^9 + 8\theta + 4\theta^2 + 3\theta^3 + 2\theta^4 + \theta^5 + 7\theta^4 + 6\theta^5 + 5\theta^6 + 4\theta^7 + 3\theta^8 + 2\theta^9 + \theta^{10} + 14\theta(1-\theta)^2(1+\theta^2+\theta^4) = 0.$$

The first line is $= 14\theta(3-6\theta+6\theta^2-6\theta^3+6\theta^4-6\theta^5+5\theta^6)$; or throwing out the factor $(1-\theta)$, the equation is

$$3-5\theta+6\theta^2-(1-\theta)^7+2(1-\theta)(1+\theta)2(1-\theta+\theta^2)(1+\theta)^2=0;$$

that is,

$$(1-\theta+\theta^2)(3-5\theta+6\theta^2)(1-\theta)^7-2(1-\theta+\theta^2)^2=0;$$

or throwing out the factor $1-\theta+\theta^2$, the equation is

$$(3-5\theta+6\theta^2)-2(1-\theta+\theta^2)=0,$$

which is an identity.

The other equation is

$$\gamma^2 + 2\beta\gamma + 2\beta\frac{u^3}{v} = u^4v^2\left(2\gamma + 2\beta\gamma + 2\frac{u^3}{v}\beta^2\right);$$

that is,

$$\gamma^2 + 2\beta\gamma + u^3v\beta^2 + 2(1+\beta)\frac{u^3}{v}(M-\gamma v^2) - 2\beta v = 0,$$

which might also be verified, but I have not done this.

[Book p. 152]

47. The conclusion is

$$\alpha = 1, \quad \beta = \frac{1}{2}\left(\frac{1}{M} - 1\right), \quad \gamma = \frac{1}{2}u^3v^2\left(\frac{1}{M} - \frac{u^7}{v^7}\right), \quad \delta = \frac{u^3}{v};$$

where

$$\frac{1}{M} = \frac{-7u(1-uv)(1-uv+u^2v^2)}{u-v^7},$$

and of course

$$\frac{1-y}{1+y} = \frac{1-x}{1+x} \left(\frac{1-\beta x + \gamma x^2 - \delta x^3}{1+\beta x + \gamma x^2 + \delta x^3} \right)^2;$$

but the resulting form may admit of simplification.

48. The endecadic transformation, $n = 11$.

I have not completed the solution, but the results, so far as I have obtained them, are interesting. The coefficients are $\alpha, \beta, \gamma, \delta, \epsilon, \zeta$; and we have, in general,

$$\alpha = 1, \quad 2\epsilon = u^5 v^2 \left(\frac{1}{M} - \frac{u^{11}}{v^{11}} \right),$$

$$2\beta = \frac{1}{M} - 1, \quad \zeta = \frac{u^5}{v}.$$

The unused equations then are

$$u^4(2\alpha\gamma + 2\alpha\beta + \beta^2) = v^4(\epsilon^2 + 2\epsilon\zeta + 2\delta\zeta),$$

$$u^4(2\alpha\delta + 2\alpha\gamma + 2\beta\gamma + 2\beta\delta) = v^4(2\delta\epsilon + 2\delta\zeta + 2\gamma\epsilon + 2\gamma\zeta + \beta^2),$$

$$u^4(2\beta + 2\beta\gamma + 2\beta\delta + 2\delta\delta + \gamma^2) = v^4(2\alpha\zeta + 2\delta\delta + 2\delta\zeta + 2\beta\zeta + \epsilon^2),$$

which, eliminating therein for $\alpha, \beta, \epsilon, \zeta$ the foregoing values, become two equations, from which, eliminating M , we should obtain the modular equation, and then M in terms of u, v .

49. To complete the calculation, we have

$$QB - AS = -uv^4 \left(\frac{v^4}{u^{16}} - \frac{u^5}{v^3} \beta^2 \right) - \left\{ u^4 \beta (2 + \beta) + \frac{v^4}{u^2} \left(\epsilon + 2 \frac{u^{10}}{v} \right) \right\} \left\{ \frac{1}{uv^2} - \frac{u^5}{v^3} + \frac{\epsilon}{u^4 v^2} \right\};$$

viz. multiplying by 8, and substituting for 2β , the their values, this is

$$8(QB - AS) = -2uv^4 \left\{ u^4 v^2 \left(\frac{1}{M} - \frac{u^{11}}{v^{11}} \right) - \frac{u^5}{v^4} \left(\frac{1}{M} - 1 \right)^2 \right\}$$

$$- \left\{ u^4 \left(\frac{1}{M} - 1 \right) \left(\frac{1}{M} + 3 \right) - u^4 v^2 \left(\frac{1}{M} - \frac{u^{10}}{v^{11}} \right) \right\} \left\{ \left(\frac{1}{M} - 1 \right) \left(\frac{1}{M} + 3v \right) + \frac{4u^{10}}{v^4} \right\},$$

or, what is the same thing

$$-\frac{8v}{u^2} (QB - AS) = 2 \left\{ u^4 v^2 \left(\frac{1}{M} - \frac{u^{11}}{v^{11}} \right) - v^4 \left(\frac{1}{M} - 1 \right)^2 \right\}$$

$$+ \left\{ \left(\frac{1}{M} - 1 \right) \left(\frac{1}{M} + 3 \right) - v^2 \left(\frac{1}{M} - \frac{u^{10}}{v^{11}} \right) \right\} \left\{ \left(\frac{1}{M} + 3 \right) \left(\frac{1}{M} - \frac{u^{10}}{v^{11}} \right) - \frac{1}{v^2} \left(\frac{1}{M} + 3 \right) \right\},$$

or, say we have $-\frac{8v}{u^2} (QB - AS) = \Pi$, where

$$\Pi = \frac{1}{M} v^2 (v - v^4)$$

$$+ \frac{1}{M} v^4 (u - 2v^3) (1 - v^4)$$

$$+ \frac{1}{M} [4u^2 v^2 + v^4 (3 - v^4) - 4uv^2 + 2u^4]$$

$$+ \dots - 2v^4 + 6v^2 v^3 (1 - v^4) + u^4 (-3 + 5v^4);$$

[Book p. 153]

where the terms containing $\frac{1}{M}$ disappear of themselves, viz. this is

$$= \frac{1}{v} \left(\frac{u^{10}}{v^{10}} - 2u^4 v^2 + 2u^3 - u^4 v^4 \right)$$

$$= -\frac{1}{v} \frac{u^4}{v^2} (v^2 + 2u^3v^4 - 2uv - u^4);$$

observe that the term in () is $= 0$, the modular equation for the case $n = 3$. As the values γ and δ appear here as fractions, having in their denominator the function $u^2 + 2ux - 2u^3v - v^4$, where

[Book p. 154]

wherefore the value of 2γ is $= \frac{1}{4} \Pi \div v^4(1 - v^4) + (u^4 + 2u^3v - 2uv - v^4)$. Similarly, writing

$$\begin{aligned} \Pi' &= \frac{1}{M'} v^4 (1 - v^4) \\ &+ \frac{1}{M'} u^4 (u^4 - 2uv^4) (1 - v^4) \\ &+ \frac{1}{M} [4u^4v^2 + u^4(3 - v^4) + 4uv + 2u^4] \\ &+ \dots - 2u^4 + 6u^2v^3(1 - v^4) + u^4(-3 + 5v^4), \end{aligned}$$

we find

$$2\delta = \frac{1}{u} \Pi' \div (v^4 + 2u^3v^4 - 2uv - u^4);$$

in verification whereof observe that this being so, the first equation gives the identity

$$\left\{ \left(\frac{1}{M} - 1 \right) \left(\frac{1}{M} + 3 \right) - v^2 \left(\frac{1}{M} - \frac{u^{10}}{v^{11}} \right) \right\} (u^4 + 2u^3v - 2uv - v^4) + \Pi - \Pi' = 0.$$

50. The result is that, writing for the moment $v^4 + 2u^3v^4 - 2uv - u^4 \equiv \Delta$, the values of the coefficients are

$$\begin{aligned} &\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \\ &= 1, \quad \frac{1}{2} \left(\frac{1}{M} - 1 \right), \quad \frac{1}{4} \frac{\Pi}{\Delta}, \quad \frac{1}{4} \frac{\Pi'}{\Delta}, \quad \frac{1}{2} u^5 v^2 \left(\frac{1}{M} - \frac{u^{11}}{v^{11}} \right), \quad \frac{u^5}{v}, \end{aligned}$$

and

$$\frac{1-y}{1+y} = \frac{1-x}{1+x} \left(\frac{A - Bx + Cx^2 - Dx^3 + Ex^4 - Zx^5}{A + Bx + Cx^2 + Dx^3 + Ex^4 + Zx^5} \right)^2;$$

the modular equation is known, and to complete the solution we require now an expression of M in terms of u, v .

51. We may herein illustrate the following theorem, viz. we may simultaneously change

$$u, v, \frac{1}{M}, \alpha; \beta; \gamma; \delta; \epsilon; \zeta; \text{ into } \frac{1}{u}, \frac{1}{v}, \frac{v^{11}}{u^{11}M}, \zeta; \epsilon; \delta; \gamma; \beta; \alpha.$$

Thus making the change in the equation

$$\frac{\beta}{\alpha} = \frac{1}{2} \left(\frac{1}{M} - 1 \right),$$

we have

$$\frac{\epsilon}{\zeta} = \frac{1}{2} \left(\frac{v^{11}}{u^{11}M} - 1 \right); \quad \text{that is, } \frac{1}{2} \frac{v^{11}}{u^{11}} \left(\frac{1}{M} - \frac{u^{11}}{v^{11}} \right) = \frac{1}{2} \left(\frac{v^{11}}{u^{11}M} - 1 \right),$$

which is right.

So in the equation $\frac{\gamma}{\alpha} = \frac{1}{2} \frac{\Pi}{\Delta}$, if for a moment (Π) , (Δ) are what Π , Δ become, the

[Book p. 155]

equation in $\frac{\delta}{\zeta} = \frac{1}{2} \frac{(\Pi)}{(\Delta)}$, that is, $\frac{1}{u^5} \frac{\Pi'}{\Delta} = \frac{1}{u^5} \frac{(\Pi)}{(\Delta)} \Pi'$; or $(\Pi) = \frac{1}{u^4} \frac{(\Delta)}{\Delta} \Pi'$; but obviously $\frac{(\Delta)}{\Delta} = -\frac{1}{u^4v^4}$; and the equation thus is

$$(\Pi) = -\frac{1}{u^4v^4} \Pi', \quad \text{or say } u^4v^4 (\Pi) = -\Pi'; \quad \text{that is,}$$

$$\begin{aligned}
 -\Pi' = u^4 v^4 & \left\{ \frac{v^{11}}{u^{11}M} \cdot \frac{1}{v^4} \left(1 - \frac{1}{v^4} \right) \right. \\
 & + \frac{v^{11}}{u^{11}M} \cdot \frac{1}{u^4} \left(\frac{1}{u^4} - \frac{2}{v} \right) \left(1 - \frac{1}{v^4} \right) \\
 & + \frac{v^{11}}{u^{11}M} \cdot \frac{4}{u^2 v^2} \left(1 - \frac{1}{v^4} \right) \left(\frac{1}{u^4} + \frac{3v^{11}}{u^{11}} \right) \left(1 - \frac{4}{v^2} + \frac{2}{u^4} \right) \\
 & \left. - \frac{2}{v^4} + \frac{6}{u^2 v^4} \left(1 - \frac{1}{v^4} \right) + \frac{1}{u^4} \left(-3 + \frac{5}{v^4} \right) \right\},
 \end{aligned}$$

which is right.

The general theory by q -transcendents. Art. Nos. 52 to 71.

52. I recur to the formula

$$\frac{1-y}{1+y} = \frac{1-x}{1+x} \left(\frac{\alpha - \beta x + \gamma x^2 - \dots \pm \sigma x^{\frac{1}{2}(n-1)}}{\alpha + \beta x + \gamma x^2 + \dots + \sigma x^{\frac{1}{2}(n-1)}} \right)^2,$$

and seek to express the ratios $\alpha : \beta : \dots : \sigma$ in terms of q . Writing with Jacobi

$$\omega = \frac{mK + m'iK'}{n},$$

we have in general

$$\alpha + \beta x + \gamma x^2 + \dots + \sigma x^{\frac{1}{2}(n-1)} = \alpha \left(1 + \frac{x}{\sin 2\omega} \right) \left(1 + \frac{x}{\sin 4\omega} \right) \dots \left(1 + \frac{x}{\sin(n-1)\omega} \right);$$

$$(\sin \omega = \sin \omega \text{ am}; \text{ viz. } \sin 2\omega = \sin(K - 2\omega), \text{ etc.});$$

and the values of $\alpha, \beta, \dots, \theta$ which correspond to the moduli $v, u, \dots, v_*$ or say the values $(\alpha, \beta_1, \dots, \theta_1), (\alpha, \beta_2, \dots, \theta_2), (\alpha_n, \beta_n, \dots, \theta_n)$ are obtained by giving to ω the values

$$\begin{aligned}
 & \omega_1, \quad \omega_2, \quad \omega_3, \quad \dots, \quad \omega_n, \\
 & = \frac{2K}{n}, \quad \frac{2K + iK'}{n}, \quad \frac{4K + iK'}{n}, \quad \dots, \quad \frac{iK'}{n},
 \end{aligned}$$

viz. the cases ω_1, ω_n correspond to Jacobi's first and second real transformations, and the others to the imaginary transformations.

I remark that $\omega = \omega_1$ gives for $\sin 2p\omega$ an expression which is rational as regards q , but $\omega = \omega_n$ gives an expression involving $q^{\frac{1}{n}}$, the real n th root of q ; the other values $\omega_2, \omega_3, \dots$ give the like expressions, involving $\alpha q^{\frac{1}{n}}, \alpha^2 q^{\frac{1}{n}}, \dots$ (α an imaginary n th root of unity, the imaginary n th roots of q).